

2. The table below gives data about some objects in our solar system.

Object	Mass (units)	Diameter (km)	Length of day (hours)	Year length (days)	Orbital speed (km/s)	Mean temperature (°C)	Distance from Sun (units)
Mercury	0.330	4 879	4 222.6	88	47.4	167	0.39
Venus	4.87	12 104	2 802	225	35	464	0.72
Earth	5.97	12 756	24	365	29.8	15	1.00
Moon	0.073	3 475	708.7		1	−20	
Mars	0.642	6 792	24.7	687	24.1	−65	1.52
Jupiter	1 898	142 984	9.9	4 331	13.1	−110	5.20
Saturn	568	120 536	10.7	10 747	9.7	−140	9.54
Uranus	86.8	51 118	17.2	30 589	6.8	−195	19.18
Neptune	102	49 528	16.1	59 800	5.4	−200	30.06
Pluto	0.0146	2 370	163.3	90 560	4.7	−225	39.53

(a) (i) Use the information from the table to **tick** (✓) the **two** correct statements below. [2]

- Neptune is hotter than the Moon.☐
- The mean temperature of the Moon is 5 degrees less than the Earth.☐
- A year on Earth is about 4 times longer than a year on Mercury.☐
- Mercury orbits the Sun with a speed around 10 times greater than Pluto.☐

(ii) Pluto was once considered to be a planet but is now classed as a dwarf planet.
Use the data in the table to suggest a reason for the change. [1]

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(iii) Ceres is another dwarf planet and it is the only dwarf planet located in the asteroid belt. Estimate its distance from the Sun. [1]

Distance from the Sun = units

(b) Elin concludes that rocky planets with the greatest mass have the shortest day length. Explain the extent to which the data supports this conclusion. [2]

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- (c) Use an equation from page 2 to calculate the velocity of the skateboarder after he catches the rugby ball. [2]

Velocity = m/s

- (d) Tomos suggests that **kinetic energy** is conserved in this collision. Explain, by using calculations, whether or not you agree. Use an equation from page 2. [4]

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END OF PAPER

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